

1.What is embeddings ?

“ Embedding is the process of converting Tokenizers into numerical vector (mathematical representation) for understating machine ”.

- Real-world objects like words, images, or audio that machine learning model can process .
- The "distance" between vectors captures semantic relationships.

2.why are embeddings needed in language model ?

Reason	Explanation
Machine-readable format	LLMs operate on numbers, not raw text—tokens must become vectors cloudflare+1
Semantic meaning	Close vectors = similar meaning, enabling the model to understand context and intent quantumblack.medium+1
Mathematical operations	Embeddings enable linear algebra and ML techniques (like transformers) to work on language quantumblack.medium
Syntactic & semantic capture	They capture relationships between word parts and solve word-sense disambiguation ceur-ws

3.How do embedding differ from tokenizers ?

Aspect	Tokenizer	Embedding
Purpose	Text preprocessing; converts text to numerical sequences medium	Creates word embeddings from integer-encoded sequences medium

Output	Arbitrary but fixed integer IDs for easy lookup linkedin	Continuous vector representations capturing meaning cloudflare+1
What it encodes	Just identification (which word/subword) linkedin	Semantic meaning and relationships between words lightly
Stage	First step: text → integers medium	Second step: integers → vectors medium

5. Real - world applications of embeddings ?

- 1.semantic search : finding documentation / items by meaning ,not just keywords .
- 2.recommendation system : youtube ,shopping platforms recommend similar content.
3. Clustering : grouping similar items together
4. RAG (retrieval augmented generation) : LLMs retrieve relevant context using embeddings